

Cs304 Object Oriented Programming

***Most Important & Repeated
Short & long Question***



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Question No 4.

Can we create an array of objects for a class having user defined constructor? Justify your answer.

Answer:-

There must always be a default constructor if we want to create array of objects

Question No 5

Define static data members.....2.mark

Answer:-

"A variable that is part of a class, yet is not part of any object of that class, is called static data member"

Question No 6.

What is Object-Orientation?

Answer:-

It is a technique in which we visualize our programming problems in the form of objects and their interactions as happens in real life.

Question No 10.

What is an abstract class?

Answer:-

Taking part of the objects and their functions that is related to the problem and ignore all other irrelative characteristic called abstraction

Question No 11.

Define Sub- typing?

Answer:-

Sub-typing means that derived class is behaviourally compatible with the Base class

Question No 12.

Define abstract Class and Concrete Class?

Answer.

These are abstract concepts and the classes we make against abstract concepts are called abstract classes.

The entities that actually we see in our real world are called concrete objects and classes made against these objects are called concrete classes.

Question No 13.

Define Association and kinds?

Answer:-

Interaction of different objects in OO model (or in problem domain) is known as association.

Kinds of Association:

There are two main types of association

1.Class Association 2.Object Association

Question No 14.

Explain two benefits of setter functions.

Answer:-

They help to define the variable in an interface

To make a variable read-only

notify other methods / classes when the value is changed

Question No 15.

Define Multiple Inheritance

Answer:-

Sometimes we want to reuse characteristics of more than one parent class, in that case we need to inherit a class from more than one classes.

Question No 19.

Define Member Function?

Answer.

We declare member function inside the class definition and declare them outside the class.

Question No 20.

Define inline function?

Answer.

Inline functions is a way used by compilers to improve efficiency of the program

Question No 21.

Define Constructor?

Answer.

Constructor is used to initialize the objects of a class

Question No 22.

Define Destructor ?

Answer.

Destructor is used to free memory that is allocated through dynamic allocation.

Question No 26.

Define Friend Functions?

The functions which are not member functions of the class yet they can access all Private members of the class are called friend functions.

Define Inheritance?

Answer:-

may have its own unique characteristics.

Answer:-

The parent class is called base class and the child class is called derived class

Question no 29.

What is interface give two real life examples?

Answer:-

Interface is a set of functions of an object that he wants to expose to other objects.

Examples

Interface of a Car is set of these functions

• Accelerate • Change Gear • Apply Brakes • Turn Lights On/Off

Interface of a Phone is set of these functions

• Input Number • Place Call • Disconnect Call • Add number to address book • Remove number • Update number

Question No30.

What is aggregation give example?

Answer:-

An object may contain a collection (aggregate) of other objects, the relationship between the container and the contained object is called aggregation, For Example: A room and its components like table, chair and bed etc are example of aggregation.

Question No 3.

Define Stream Insertion operator and Stream Extraction operator ?

Answer.

Stream Insertion operator

To display the data on the screen c++ provides us insertion operator ('<<') to put Data on output stream.

Stream Extraction operator

C++ provided stream extraction operator ('>>') that is used to get data from Input stream.

Question No4.

Explain what type of copy the default assignment operator "=" does when applied to objects. (shallow copy or deep copy)

Answer:-

If our class has any data member using dynamic memory then we had to write our own code for default Constructor, copy constructor and similarly assignment operator as compiler generated version of these Functions performs shallow copy that creates dangling pointer, and memory leakage issues in case of dynamic memory allocation.

Question No 1.

Explain the difference between the static variable of a class with none static variable with the help of example

Answer:-

The difference between static and non static members is only that a non static member is tied to an instance of a class although a static member is tied to the class, and not to a particular instance. That is, a static member is shared by all instances of a class although a non static member exists for each instance of class.

Question No2.

What is difference between simple association and composition?

Answer:-

In Association, interacting objects have no intrinsic relationship with other object. It is the weakest link between objects. While in Composition An object may be composed of other smaller objects, the relationship between the “part” objects and the “whole”.

Question No 6.

What are binary operators? Give an example of binary operators overloading using any class.

Answer:-

Binary Operators Overloading:

Binary operators act on two quantities.

Examples of binary operators:

Examples:

Overloading + operator:

Class Complex{

Private:

Double real, img;

Public:

...

Complex operator +(const Complex &rhs);

};

Complex Complex::operator +(const Complex &rhs){

Complex t;

t.real = real + rhs.real;

t.img = img + rhs.img;

return t;

}

The binary operator is always called with reference to the left hand argument.

Question No 8.

Give c++ code to overloaded unary “—” oprators to compls member class.

Answer:-

```
Class Complex{
...
Complex operator – (int);
// friend Complex operator –(const Complex &, int);
}
Complex Complex::operator – (int){
Complex t = *this;
Real - = 1;
Return t;
}
Complex operator – (const
Complex & h, int){
Complex t = h;
h.real - = 1;
return t;
}
```


Question No 10.

Write c++ code for operator [] using string class.....5 marks.....

Answer:-

```
Class String{  
    ...  
    Public:  
    Char & operator()(int);  
    ...  
};  
Char & String::operator()  
(int pos){  
    Assert(pos>0 && pos<=size);  
    Return bufferPtr[pos-1];  
}  
Int main(){  
    String s1("Ping");  
    Char g = s1(2); // g = „i“  
    S1(2) = „o“;  
    Cout<< g << “\n”;  
    Cout<<str.GetString();  
    Return 0;  
}
```

Question No 11.

What are Accessor Functions, Explain with an example?

Answer:-

Accessor functions are used to access private data of the object, we provide accessor functions to get and set private data members of the class.

Example

```
Class Student{
```

```
...
```

```
IntrollNo;
```

```
Public:
```

```
Void setRollNo(intaRollNo){
```

```
rollNo = aRollNo;
```

```
}
```

```
};
```

Avoiding Error

```
Void Student::setRollNo(intaRollNo){
```

```
If(aRollNo < 0){
```

```
rollNo = 0;
```

```
}
```

```
Else
```

```
{
```

```
rollNo = aRollNo;
```

```
}
```

```
}
```

Question No 13.

Consider the class given below what will be the values in variables x,y and z after creating object of this

```
Class,  
Class ABC{  
Int x;  
Int y;  
Int z;  
Public:  
ABC();  
};  
ABC::ABC():x(10),z(x),y(x)  
{  
...  
}
```

Answer:-

X=10,z=10 and y=10

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